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MILITARY SPECIFICATION

FILTER ELEMENTS, LUBRICATING OIL, ENGINE, (LARGE)

This specification was approved by the Departments of the Army, the Navy, and the Air Force for use of procurement services of the respective Departments.

1. SCOPE

1.1 This specification covers oil filter elements (large) for removing foreign material (contaminants) from the oil used for lubricating internal combustion engines.

2. APPLICABLE SPECIFICATIONS, STANDARDS, DRAWINGS, AND PUBLICATIONS

2.1 The following specifications and standards, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

FEDERAL

NN-B-591 — Boxes, Fiberboard, Wood-Cleated (for Domestic Shipment).
NN-B-601 — Boxes, Wood-Cleated-Plywood, for Domestic Shipment.
NN-B-621 — Boxes, Wood, Nailed and Lock-Corner.
NN-B-631 — Boxes, Wood, Wirebound, for Domestic Shipment.
LLL-B-631 — Boxes, Fiber, Corrugated (for Domestic Shipment).
LLL-B-636 — Boxes, Fiber, Solid (for Domestic Shipment).

MILITARY

JAN-P-105 — Packaging and Packing for Overseas Shipment — Boxes, Wood, Cleated, Plywood.
JAN-P-106 — Packaging and Packing for Overseas Shipment — Boxes, Wood, Nailed.
JAN-P-116 — Packaging and Packing for Overseas Shipment—Preservation; Methods of.
JAN-P-120 — Packaging and Packing for Overseas Shipment—Cartons, Folding Paperboard.
JAN-B-121 — Barrier-Materials, Greaseproof.
JAN-P-125 — Packaging and Packing for Overseas Shipment—Barrier-Materials, Waterproof, Flexible.
JAN-P-127 — Packaging and Packing for Overseas Shipment—Tape, Adhesive, Pressure-Sensitive, Water-Resistant.
JAN-P-133 — Packaging and Packing for Overseas Shipment — Boxes, Set-Up, Paperboard.
JAN-P-139 — Packaging and Packing for Overseas Shipment—Plywood, Container-Grade.
JAN-P-140 — Packaging and Packing for Overseas Shipment—Adhesive, Water-Resistant, Caseliner.
MIL-S-901 — Shockproof - Equipment, Class HI (High-Impact): and Tests for (Shipboard Application).

NAVY DEPARTMENT

General Specifications for Inspection of Material.
14-O-13 — Oil; Lubricating (Diesel-Engine).

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STANDARDS

MILITARY

MIL–STD–105 — Sampling Procedures and Tables for Inspection by Attributes.

MIL–STD–129 —Marking of Shipments.

(Copies of specifications, standards, and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Qualification.—Filter elements furnished under this specification shall be a product which has been tested and has passed the qualification tests specified in section 4 (see 6.3).

3.2 Material.—The materials used shall be of uniform quality and composition, and shall be suitable to the current standard practices of manufacturers producing first class equipment of the kind covered by this specification.

3.3 Design.—Filter elements shall be of adequate design for the intended application (see 6.1), shall be in accordance with this specification, and shall be readily removable from the body of the installed filter without disconnecting any attached oil lines or fittings.

3.3.1 *Pull-out-device.*—The elements shall be equipped with a suitable pull-out device fixed to the top and so constructed as to withstand a direct pull of 100 pounds when tested as specified in 4.6.1.

3.3.2 *Seal gaskets.*—The elements shall be equipped with compression-type, end seal gaskets, one each at top and bottom, either mechanically secured to or an integral part of the filter elements. The gaskets shall form an effective seal when used in combination with a smooth, flat sealing plate or with each other (when elements are stacked two-high) under a sealing force of 100 pounds, and shall withstand a maximum sealing force of 500 pounds. The seals shall be made of a material that will not shrink or disintegrate due to long periods of immersion in hot lubricating oil.

3.4 Dimensions.—Filter elements shall be of the following dimensions:

	Inches
Length, end to end of sealing gaskets	18 ± 1/16.
Outside diameter (max.)	7 5/16.
Inside diameter of center tube or centering device	2 1/32 to 2 3/32.
Inside diameter of gasket (max.)	3.
Outside diameter of gasket (min.)	4.
Width of sealing gasket (min.)	1/2.

3.5 Shockproofness.—Filter elements shall withstand the shockproof test specified in 4.6.2.

3.6 Flow rate.—When determined by the method specified in 4.7.1, the flow rate through any elements shall not vary more than plus or minus 20 percent from the rate as specified in 4.6.3.

3.7 Abrasive content. — The abrasive content of the lubricating oil shall not exceed 0.01 percent by weight when tested as specified in 4.6.4. For the purpose of determining abrasive content, the oil sample shall be taken from the system on the intake side of the filter.

3.8 Additive removal.—The element shall not remove more than 30 percent of any additive agents from any Diesel engine lubricating oils conforming to Specification 14–0–13. The test for additive removal shall be made as specified in 4.6.5.

3.9 Media migration. — The filter element shall be so constructed that none of the filtering media shall be discharged into the engine oil stream (see 4.6.6), or be dissolved by the lubricating oil, or in any way deleteriously affect the lubricating oil.

3.10 Pressure drop.—After subjection to the tests specified in 4.6.2 and 4.6.4, the pres-

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sure drop across one element shall be not greater than 30 p.s.i. from that determined by the test specified in 4.6.3 at a flow rate corresponding to a pressure drop of 15 p.s.i.

3.11 Workmanship. — The workmanship shall be first-class in every respect.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Inspection procedures. — For Naval purchases, the general inspection procedures shall be in accordance with General Specifications for Inspection of Material.

4.2 Qualification tests at a laboratory.— Qualification tests shall be conducted at a laboratory designated by the Bureau of Ships. These tests shall consist of the tests specified in 4.6.

4.3 Inspection lot.—All elements presented at one time shall be considered a lot for purposes of inspection and tests. The maximum number of elements in any inspection lot shall be limited to 3,000.

4.4 Sampling for lot acceptance.

4.4.1 *Sampling for inspection.*—A random sample of elements shall be selected from each inspection lot of material offered for Government inspection of visual and dimensional characteristics, with lot acceptance based on the sampling inspection requirements shown in table I, in accordance with Standard MIL–STD–105.

4.4.2 *Sampling for production check tests.* —From each inspection lot, the Government inspector shall select a random sample of one element for the tests specified in 4.7. The sample so selected shall be taken from the sample units subjected to the lot acceptance inspection of 4.5.1 and found satisfactory. The sample shall be forwarded to a Government laboratory designated by the bureau or agency concerned.

TABLE I.—*Sampling for visual and dimensional inspection AQL (approx.) = 1.5 percent defective.*

Number of elements in inspection lot	Number of elements in sample	Acceptance number (defectives)	Rejection number (defectives)
15 and under	7	0	1
16 to 40	10	0	1
41 to 110	15	0	1
111 to 300	25	1	2
301 to 500	35	1	2
501 to 800	50	2	3
801 to 1300	75	3	4
1301 to 3000	110	4	5

4.5 Lot acceptance inspection (at place of manufacture).

4.5.1 *Visual and dimensional inspection.*— Each of the sample elements selected in accordance with table I shall be inspected by the Government inspector to determine conformance with this specification, and to determine that the design, weight, dimensions, and details of construction are essentially equivalent to those of the approved element. Any element in the sample containing one or more visual or dimensional defects shall be rejected, and if the number of defective elements in any sample exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected. Rejected lots may be offered again for Government inspection provided the contractor has removed all nonconforming elements. The Government inspector shall again select and examine samples from such rejected lots to verify compliance with this specification.

4.6 Test procedures.

4.6.1 *Pull-out device test.*—With the media container of the sample held stationary, the pull-out device shall be subjected to a straight (tension) pull gradually increased to a maximum of 100 pounds. During this test the device and its attaching means shall be observed for evidence of deformation or failure.

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4.6.2 *Shockproof test.* — Filter elements shall be tested in accordance with type C of Specification MIL–S–901.

4.6.3 *Flow rate calibration.*—One element shall be tested to determine the flow rates corresponding to pressure drops of 5, 10, 15, 20, and 30 p.s.i. across the element using 170 seconds Saybolt Universal viscosity, clean lubricating oil.

4.6.4 *Filtering performance.* — Filtering performance shall be determined by connecting a filter containing one element into a recirculating oil system and measuring the capacity of the element to remove contaminants from the lubricating oil. The test shall be made under the following conditions:

Type of oil used (see Spec. 14–O–13, Navy Symbol 9250).	
Viscosity of oil during test.	170 (Saybolt Universal) second.
Quantity of oil in system.	16 gallons.
Pressure drop across filter.	20 p.s.i.
Rate of oil flow through filter. (include flow through element and relief valve).	2.0 g.p.m.
Rate of addition of contaminants:	
Carbon black (average particle size below 0.00004 inch).	0.00020 pound per minute.
Abrasive (particle size, average 0.0003 to 0.0007 inch, 80 percent shall be between 0.001 and 0.0001 inch).	0.000020 pound per minute.
Carbon tetrachloride soluble, hexane insoluble asphaltenes, softening point between 130° and 180° F.	0.00005 pound per minute.
Duration of test	200 hours.
Addition of contaminants.	Continuous at a constant rate using clean oil at start.
Mounting of filter, body.	On a vibrating base having a vertical amplitude of vibration of not less than 0.015 inch and a vibrating frequency of not less than 3,600 cycles per minute.

4.6.5 *Additive removal test.*—Eleven gallons of clean, additive-type lubricating oil symbol 9250 conforming to Specification 14–O–13 shall be recirculated through the element at a flow rate of 2 g.p.m. for 100 hours at 170° F. Samples shall be obtained at 25-hour intervals and analyzed for additives content. Not less than two test runs shall be made using a new filter element and a different oil containing a different type of additive for each run.

4.6.6 *Filtering media migration test.*—The element shall be installed in a standard body containing an internal relief valve set to open at 25 p.s.i. and mounted in front of and in series with an 0.0015-inch spacing metal-edge strainer. Navy symbol 9250 oil, conforming to Specification 14–O–13, contaminated with 0.5 percent tap water, shall be circulated through the element at a pressure drop of 30 p.s.i. (across element and body) and a temperature of 160° F., for 200 hours. The metal-edge strainer shall be examined after 10 hours and every 50 hours thereafter for evidence of the filtering media. After the 200-hour test, a complete examination shall be made of the test element, lubricating oil, and metal-edge strainer.

4.7 **Production check tests (at a Government laboratory).**—Inspection tests shall be made at a Government laboratory on the sample element selected as specified in 4.4.2. The tests shall consist of surface inspection and the tests specified in 4.7.1 and 4.7.2, to determine whether the element is equal in quality to elements approved for Government use under previous qualification tests.

4.7.1 *Flow rate vs. pressure drop.*—Flow rate vs. pressure drop shall be determined by circulating clean lubricating oil having a viscosity of 170 seconds Saybolt Universal through the element at a pressure drop of 20 p.s.i. across the element and determining the corresponding flow rate.

4.7.2 *Weights and dimensions.* — The weights and dimensions of all filter elements submitted for inspection shall be checked for

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comparison with weight and dimensions of the elements submitted for qualification approval.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging for domestic and overseas shipment.

5.1.1 Filter elements shall be individually preserved in accordance with method 1A-2 of Specification JAN-P-116, using type P-10 preservative on metal parts only; shall be wrapped with barrier-material conforming to grade A of Specification JAN-B-121 and sealed with water-resistant tape conforming to Specification JAN-P-127.

5.1.2 Filter elements shall be packaged in folding paperboard cartons, set-up paperboard, corrugated fiber, or solid fiber boxes, conforming to Specification JAN-P-120, JAN-P-133, LLL-B-631, or LLL-B-636, respectively.

5.2 Packing.

5.2.1 *For domestic shipment.* — The filter elements shall be packed in accordance with 5.2.1.1 or 5.2.1.2, as specified (see 6.1).

5.2.1.1 Filter elements packaged as specified in 5.1, shall be packed in substantial commercial containers of the type, size and kind commonly used for the purpose, in such a manner as to insure acceptance and safe delivery to the designated point by common or other carriers, at the lowest rate of the carrier.

5.2.1.2 Filter elements, packaged as specified in 5.1, shall be packed in wood-cleated-fiberboard, wood-cleated-plywood, nailed wood, wirebound wood, corrugated fiber, or solid fiber boxes conforming to Specification NN-B-591, NN-B-601, NN-B-621, NN-B-631, LLL-B-631, or LLL-B-636, respectively. Gross weight of container and contents shall not exceed 90 pounds.

5.2.2 *For overseas shipment.*—Filter elements, packaged as specified in 5.1, shall be packed in cleated plywood boxes conforming to style A or B of Specification JAN-P-105 or nailed wood boxes conforming to style 2, 2½, or 3 of Specification JAN-P-106. Plywood shall conform to type A or B of Specification JAN-P-139. Boxes shall be strapped in accordance with the applicable box specification. Gross weight of container and contents shall not exceed 90 pounds.

5.2.2.1 Shipping containers for overseas shipment shall be furnished with a sealed case liner made from material conforming to Specification JAN-P-125. Seams and closures shall be made with adhesive conforming to Specification JAN-P-140, and shall have water-resistance equal to that of the body material.

5.3 Marking.—In addition to any special marking required by the contract or order, marking shall be in accordance with Standard MIL-STD-129.

6. NOTES

6.1 Intended use. — Lubricating oil filter elements are intended for use with lubricating oil filters conforming to Specification GGF5.

6.2 Ordering data. — Procurement documents should specify the following:

(a) Title, number and date of this specification.

(b) Whether packing for domestic or overseas shipment is required, and if domestic, the type required (see 5.2).

6.3 In the procurement of products requiring qualification the right is reserved to reject bids on products that have not been subjected to the required tests and found satisfactory for inclusion on the Military Qualified Products List. The attention of suppliers is called to this requirement, and manufacturers are urged to communicate with the Bureau of

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